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Power System Operation - Lab

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WiSe 2023/2024

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Session 3

Exercise 9-12

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Exercise 9

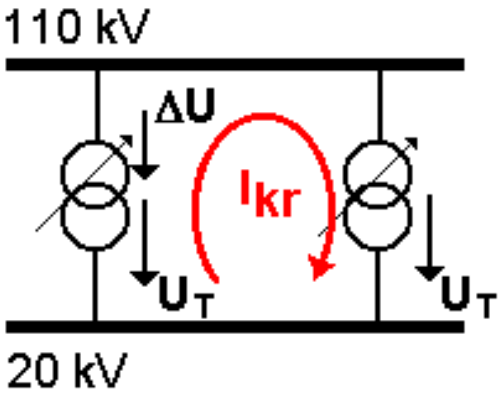
Parallel transformers and circulating currents

- The transformers TR 69 and 70 in „WEST 20“ are constructed identically.
- Turn both TR 69 and TR 70 in system „WEST 20“ to the same step and both voltage controllers to off („AUS“). Please note the load side values:

	Stufe	U [kV]	P [MW]	Q [MVAr]
Trafo 69				
Trafo 70				

- Change the step changer values by -4 for TR 69 and by +4 for TR 70

	Stufe	U [kV]	P [MW]	Q [MVAr]
Trafo 69				
Trafo 70				



- Explanation of the values: The set voltage difference (different numbers of steps) drives an additional (reactive) circulating current I_{Kr}

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Exercise 10

Fault scenario: Parallel transformers and regulator fault

- In the “West 20” plant, switch the two transformers in parallel at the same level and both voltage regulators "ON".
- In this situation, the regulator suffers a malfunction.
- Have the supervisor (from the trainer's station) bring it in:
  - At first, just observe what happens
  - Consider how to prevent the events from escalating
  - Consider how to return the network situation to a normal state at the end of the events
  - Consider how to break the chain of events.

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Exercise 11

Ground fault in the line network of the “West 20” plant

- Observe the 3 conductor voltages in “West 20”.
- Then have the trainer bring in the ground fault.
- In the “West 20” plant, the ground fault relays respond.
- First observe what happens.
  - Again, observe the 3 line voltages.
  - Consider where the fault may be.
  - Consider how to locate and clear the fault without disconnecting customers.
  - Discuss troubleshooting strategies.
  - Initiate the actions.
  - Perform the switching in a coordinated manner with the route service.
  - Conduct the switching calls.

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Exercise 12

a) Short circuit in the route network

- Have the short-circuit brought in by the supervisor.
- The protective relays respond. Also note the protocol
  - Consider where the fault may be and how to locate and clear the fault location.
  - Discuss troubleshooting strategies.
  - Initiate the actions to be taken.
  - Execute the circuits in a coordinated manner with the track service.
  - Attention: never(!) let the line staff switch to short-circuit (our switch-disconnectors are not short-circuit proof).
  - Practice the switching talks

b) Short circuit in the meshed 400 kV network

- Select a short-circuit location in the 400 kV network with a supervisor. Discuss in advance the reactions of the distance protection relays.
- Have the short-circuit played in by the supervisor.
- The protection relays respond. Observe the protection pick-ups and cut-outs.
- Has the fault been selectively de-energized?
- Determine the relay run times

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Free Discussion

Further tasks/errors/problems/solutions and discussions

- Network switching with load flow optimization
- Track and improve yield situation
- Yield optimized grid operation
- Switching island grids and saving them from overloads
- Frequency behaviour in islanding grid
- Short-circuit analysis in the transmission network and protection selectivity
- Track CENS indicators (penalties) in the event of a fault